Automatic Category Suggestion for Wikimedia Commons

Team Number: 72

The Problem:

- Commons has millions of files, and volunteers categorize them by hand.
- Labels are incomplete and inconsistent, and new files arrive every day.
- Our goal is to suggest likely categories and let a human decide what suits the best.

Our Understanding

The challenge	Our response
Millions of categories, always changing	Retrieve similar images instead of a fixed classifier
A missing label isn't a wrong label	Treat missing as unknown, not negative
Editors use the most specific category only	Apply category-graph rules
Events and places aren't visible in pixels	Add title, date and location
Wikimedia needs open, human-reviewed tools	Explain with similar images; the editor decides

Idea and Objectives

Suggest the most likely commons categories for a new image. A human editor makes the final call.

We propose an open, explainable system that suggests Commons categories for new images by combining similar-image voting with category-name matching, with editor review, no-retraining updates, and testing on a time-based hold-out set.

- **Suggest accurate categories for new images**
  To be tested on a time-based hold-out set
- **Cover objects, scenes, places and events**
  Results planned per category type
- **Stay fast as data grows**
  Latency and index size to be measured
- **Stay robust to low-quality uploads**
  To be tested on blurred and compressed images
- **Learn new categories without retraining**
  Planned demo: add a category instantly
- **Respect Wikimedia's values**
  Open, explainable, human-reviewed

How it works:

Step	Stage	Detail	Type
1	Upload Image	+metadata	-
2	Image Encoder	CLIP	AI Step
3a	Similar Images	kNN vote	AI Step
3b	Category Names	text match	AI Step
4	Fusion	combine scores	AI Step
5	Top-k list	ranked suggestions	AI Step
6	Editor Review	accept or reject	Human Decision
7	Update Index	No retraining	-

Legend: AI Step (orange), Human Decision (blue). Steps 3a and 3b both follow the Image Encoder and both feed Fusion.

Testing Methodology

Part	What goes on the slide
Data	Open-licensed Commons files via the MediaWiki API, on a prototype subset of categories
Split	By upload date: older uploads build the index, newer ones test it, near-duplicates removed
Metrics	Precision, recall and F1 at k, plus mAP
Comparison	Each scorer alone vs the combined system
Breakdown	Common vs rare categories, by category type, and on blurred and compressed images
Fair scoring	Human review of a sample of “wrong” suggestions; report strict and reviewed scores
Efficiency	Latency, throughput and index size